



CUBAN NATIONAL CHEMISTRY CONTEST 2022

"You should never be afraid of what you're doing, when it's right."

Marie Curie

Name(s): _____ Last name(s): _____

School: _____

Municipality: _____ Province: _____

CODE: _____ Test: **DAY 1. COMMON EXAM**

Instructions

- Write clearly all your data in the appropriate place (names, last names, school, municipality, province). Do not write anything in the field CODE.
- Verify that your questions booklet is complete. This is the Day 1 problem set, which has 4 problems (problem 1, 16 items; problem 2, 4 items; problem 3, 4 items; problem 4, 4 items) and a total of 6 pages.

10th grade students only need to answer questions 1, 2, and 4. 11th and 12th grade students only need to answer questions 1, 3, and 4.

- All data, constants, conversions and basic equations needed are included in the exam. You are allowed to use a scientific calculator.
- At the beginning of each problem as its scoring (from a total of 100 points), as each item scoring (in marks) is indicated.

Success!



Ministerio de Educación
República de Cuba



CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Equation of ideal gas	$p \cdot V = n \cdot R \cdot T$
Gases constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamics	$Q = \Delta E + W$
Zero on Celsius scale	273,15 K	Gibbs' free energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equation	$k = A \cdot \exp\left(-\frac{E_A}{R \cdot T}\right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$	$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$		$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$	

PERIODIC TABLE OF ELEMENTS

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1																	18
1 H 1.008	2											13	14	15	16	17	2 He 4.003
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3	4	5	6	7	8	9	10	11	12	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.63	33 As 74.92	34 Se 78.97	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.95	43 Tc -	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57-71	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po -	85 At -	86 Rn -
87 Fr -	88 Ra -	89-103	104 Rf -	105 Db -	106 Sg -	107 Bh -	108 Hs -	109 Mt -	110 Ds -	111 Rg -	112 Cn -	113 Nh -	114 Fl -	115 Mc -	116 Lv -	117 Ts -	118 Og -
57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm -	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0			
89 Ac -	90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np -	94 Pu -	95 Am -	96 Cm -	97 Bk -	98 Cf -	99 Es -	100 Fm -	101 Md -	102 No -	103 Lr -			

Problem 1. Miscellanea (16 points)

1.1	...	Total	Total
1	idem	16	16

Select the right answer in each case.

1.1. In the sequence of ions S^{2-} , Br^- , Ag^+ , Cu^{2+} , Mg^{2+} , the one with the largest ionic radius is:

A) S^{2-}	B) Ag^+	C) Br^-	D) Cu^{2+}
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1.2. Group of chemical species with the same bond order:

A) NO^+ , O_2^{2+} , O_2^- , N_2 , CN^-	B) N_2^{2-} , O_2^{2-} , NO^- , O_2 , CN^+
C) NO , O_2^+ , N_2^- , NO^{2+} , CN	D) none of the previous

1.3. Which molecule is polar and has a sp^2 -hybridized central atom?

A) BCl_3	B) $HClO_2$	C) $POCl_3$	D) H_2CO_3
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1.4. The covalent character of the bond present in the halides $LiCl$, $BeCl_2$, BCl_3 and CCl_4 increases in the order:

A) $LiCl < BeCl_2 < BCl_3 < CCl_4$	B) $CCl_4 < BCl_3 < BeCl_2 < LiCl$
C) $LiCl < BeCl_2 < BCl_3 > CCl_4$	D) $LiCl < BCl_3 < BeCl_2 < CCl_4$

1.5. Which molecule can act as a Lewis acid?

A) $SnCl_4$	B) SiF_4	C) CCl_4	D) SF_4
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1.6. If the enthalpy of formation of liquid water is -286 kJ/mol and that of ice is -292 kJ/mol , how much energy is released in the combustion of 1 g of ice?

A) 6 kJ	B) 16.2 kJ	C) 0	D) 333 J
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1.7. Which oxide has amphoteric acid-base behavior?

A) CaO	B) ZnO	C) CO_2	D) SiO_2
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1.8. The bromination of acetone varies its initial rate from $7.6 \text{ mol/(L}\cdot\text{s)}$ to $15.2 \text{ mol/(L}\cdot\text{s)}$ when pH decreases from 1.3 to 1, keeping constant the concentrations of the other reagents. Knowing that $pH = -\log(H^+)$, what is the reaction order respect to H^+ ?

A) -1	B) 1	C) 0,5	D) 2
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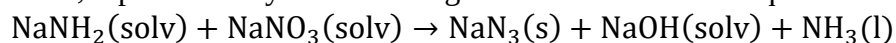
1.9. Which of the following chemical species is paramagnetic?

A) H_2	B) HHe^+	C) He_2^+	D) He_2
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1.10. In which of the oxo-acids below the oxidation state of phosphorus isn't 5+?

A) $H_4P_2O_7$	B) H_3PO_4	C) $H_4P_2O_6$	D) $H_5P_3O_{10}$
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1.11. In the reaction between sodium amide and sodium nitrate, in liquid ammonia under controlled conditions, represented by the following unbalanced chemical equation:



for every mole of sodium amide that is consumed, the following amount is formed:

A) 2 mol of NaN_3	B) 1 mol of NaN_3
C) 1/2 mol of NaN_3	D) 1/3 mol of NaN_3

1.12. Which molecule has T-shaped geometry?

A) $SOCl_2$	B) ICl_3	C) PBr_3	D) $COCl_2$
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1.13. For the reaction between Cl_2 and NO , producing NOCl , it is known that at low concentrations of the reducing agent, the experimental rate law reduces to $v = k \cdot c^2(\text{NO}) \cdot c(\text{Cl}_2)$. What could be the complete expression (not simplified) that is valid for all cases?

A) $v = \frac{k_2 k_1 c^3(\text{NO}) c(\text{Cl}_2)}{k_{-1} + k_2 c(\text{NO})}$	B) $v = \frac{k_2 k_1 c^2(\text{NO}) c(\text{Cl}_2)}{k_{-1} + k_2 c(\text{NO})}$
C) $v = \frac{k_2 k_1 c^2(\text{NO}) c(\text{Cl}_2)}{k_{-1} + k_2 c(\text{Cl}_2)}$	D) $v = \frac{k_2 k_1 c^3(\text{NO}) c(\text{Cl}_2)}{k_{-1} + k_2 c(\text{Cl}_2)}$

1.14. A gas occupies a volume of 9.23 L at 345 K and 1.40 atm. The volume that this gas will occupy at 525 K and 3.20 atm will be:

A) 5.65 L	B) 6.14 L	C) 321 mL	D) 11.9 L
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1.15. Which of the following ions doesn't have a completely filled d orbital?

A) Ag^+	B) Ga^{3+}	C) Cu^{2+}	D) Zn^{2+}
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1.16. A 1.50 g sample of a silver-containing mineral was dissolved and all silver is converted to 0.124 g of Ag_2S . The percentage of silver in the ore sample was:

A) 10.8%	B) 6.41%	C) 8.27%	D) 7.20%
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Problem 2. Identify the elements (20 points)

[ONLY FOR 10th GRADE]

2.1	2.2	2.3	2.4	Total	Total
8	2	7	3	20	20

A volatile liquid with formula XY_2 is produced by passing a current of the hydride XH_4 over the simple substance of the element Y , also forming the gaseous compound with an unpleasant odor H_2Y . Combustion of 1 mol of XY_2 in an oxygen atmosphere produces 172.1 g of the oxides XO_2 and YO_2 . The mass ratio of element Y to element X in XY_2 is 5.339. Reaction of XY_2 with ammonia, in the presence of a catalyst, gives as the only products H_2Y and an ammonium salt containing 42.1% Y and a molar ratio of element X to nitrogen of 1:2.

2.1 Identify elements X and Y . Justify numerically.

2.2 Illustrate the Lewis structure of XY_2 molecule. Why does this compound have a relatively low boiling point?

2.3 Write the formula and the name of the salt obtained in the reaction with ammonia. Show the calculations that support your reasoning.

2.4 Write the balanced equations corresponding to the processes described above.

Problem 3. Bauxite (20 points)
[ONLY FOR 11th and 12th GRADES]

3.1	3.2	3.3	3.4	Total	Total
6	25	5	4	40	20

Bauxite is the most used mineral to obtain aluminum. In addition to aluminum, the main metals contained in bauxite are iron, silicon and titanium. Bauxite composition varies depending on where it is mined. In a mine of this mineral, in Greece, it was found that when water was added to the product a solid residue made up of Al_2O_3 , SiO_2 , Fe_2O_3 and TiO_2 remained undissolved. This residue, by subsequent analysis, was found to have an approximate composition $m\text{Al}_2\text{O}_3 \cdot n\text{SiO}_2 \cdot p\text{Fe}_2\text{O}_3 \cdot q\text{TiO}_2$, where occur the relationships $m/p = p/q$ and $q/n = n/p$.

The addition of 1.215 g of bauxite ore, extracted directly from the mine, to a diluted HNO_3 solution, gives a solid residue of 0.0820 g. On the other hand, by taking the same mass of bauxite and adding a NaOH solution, the mass of residue that remains at the bottom has a mass of 0.3735 g. It is known that the Fe^{3+} ion in the presence of NaOH forms a precipitate with the formula $\text{FeO}(\text{OH})$.

3.1 Write balanced equations of the reactions described. (Note that under these conditions neither SiO_2 nor TiO_2 react)

3.2 Calculate the mass composition of bauxite ore.

3.3 Propose a formula for the residue $m\text{Al}_2\text{O}_3 \cdot n\text{SiO}_2 \cdot p\text{Fe}_2\text{O}_3 \cdot q\text{TiO}_2$, taking the smallest coefficient to be equal to 1.

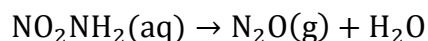
3.4 What is the maximum mass of Al that can be obtained from 1 t of this bauxite ore, assuming 85% yield in the entire metal separation and purification process.

If you could not solve part 3.2, consider $\omega(\text{Al}_2\text{O}_3) = 50\%$

Problem 4. Kinetics (14 points)

4.1	4.2	4.3	Total	Total
7	1	6	14	14

Nitramide slowly decomposes in aqueous solution according to the reaction:

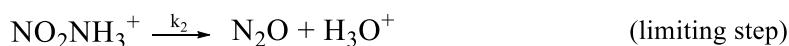
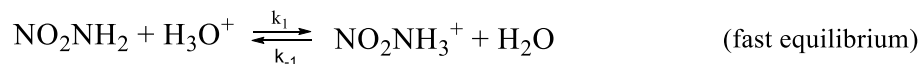


The kinetic law, determined experimentally, is the following:

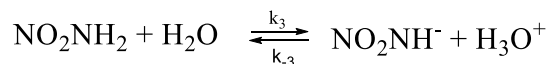
$$\frac{d(\text{N}_2\text{O})}{dt} = k \frac{[\text{NO}_2\text{NH}_2]}{[\text{H}_3\text{O}^+]}$$

4.1 Which of the following mechanisms is the most appropriate for the interpretation of the experimental kinetic law?

Mechanism 1



Mechanism 2



4.2 Find the relationship between the experimental constant and the rate constants of the individual steps in the chosen mechanism.

4.3 If the rate of the reaction is increased five-fold when increasing the temperature by 15°C from 298 K, calculate the activation energy of this reaction.